Paper 5 | NeuroInsight

An agent-navigable, modular, reproducible neuroimaging research workbench

Comprehensive research concept • critical related-work review • public tool architecture • evaluation protocols • manuscript plans • journal tiers • publication gates

Planning version: 25 October 2026 | Status: proposed research protocol; no experimental results are claimed

Prepared to support an independent, rigorous submission strategy. Software maturity, institutional permissions, benchmark outcomes and citations must be verified by authors.

Executive decision brief

NeuroInsight should be built and evaluated as a reusable research platform that joins an extensible plugin/workflow model, BIDS-aware dataset handling, reproducible local/HPC execution, quality-control and provenance, and optional agent interaction. The scientific contribution must be proven against established platforms—not inferred from the presence of a web UI or LLM.

Question	Provisional answer
Can a software paper be published?	Yes, with open code, documented reproducibility, compelling use cases, and meaningful comparisons.
What is the central user deliverable?	A dependable neuroimaging workbench with executable, independently reproducible analysis workflows.
What is the main novelty risk?	CBRAIN, brainlife.io, ChRIS, Flywheel, Nipype/Pydra, BIDS Apps, Boutiques, and Neurodesk already address large portions.
Most natural target	Imaging Neuroscience Software Toolbox or PLOS Computational Biology Software, conditional on results and fit.
Stretch target	Nature Methods only if a genuinely novel general life-science method and strong biological application are established.

Contents / research work packages

1. Concept, hypotheses and contribution boundaries

2. Critical literature and ecosystem mapping

3. NeuroInsight public product: architecture and features

4. Design contracts: BIDS, tools, compute, provenance and agent safety

5. Work packages and public release deliverables

6. Experimental plan, datasets, comparators and evaluation

7. Statistics, uncertainty, threats and reproducibility

8. Manuscript packages, figures, abstracts and review objections

9. Journal tiers and explicit submission gates

10. Timeline, resourcing, institutional and ethical controls

11. Prioritized bibliography and source links

12. Final submission readiness checklist

1. Research concept and scope

Problem. Neuroimaging analysis often requires researchers to coordinate disparate applications, formats, containers, computing environments, manual parameter entry, data movement, quality checks and provenance. Integration cost creates errors and impedes independent repeatability.

Proposed answer. NeuroInsight provides user-centered execution of reusable scientific tool plugins and multi-step workflows, with machine-readable contracts, deterministic preflight checks, explicit authorization, cross-environment execution adapters and complete execution receipts. An AI interface is optional; deterministic operation remains first-class.

Primary software/research question (Q1). Can an integrated modular platform improve reproducibility, setup burden and operational reliability compared with comparable established neuroimaging execution approaches?

Secondary question (Q2). Does a machine-readable capability layer improve task configuration and workflow discoverability without increasing invalid executions or hiding scientific assumptions?

Boundary. Paper 5 is a platform/software investigation. Formal semantic validity belongs primarily to Paper 2; agent benchmarking to Paper 3; a capability-contract algorithm belongs to Paper 1. Paper 5 may integrate those packages but cannot claim their results as a separate novel finding without independent evaluation.

- H1: Independent users complete representative workflows with lower setup time and fewer configuration errors than prespecified comparators.
- H2: Across supported deployments, configurations yield equivalent output artifacts (exact equality where appropriate; tolerance-based equivalence for numerical outputs).
- H3: Structured provenance allows an independent researcher to reconstruct runs and diagnose failures more reliably than baseline reporting.
- H4 (exploratory): Agent assistance reduces configuration effort without degrading scientific correctness when authorization and preflight checks are enforced.

2. Critical prior work: what already exists

Important: do not describe NeuroInsight as the first workflow UI, first platform for neuroimaging, first agent interface, or first reproducible containerized pipeline. The prior ecosystem is mature; the gap must be demonstrated by a feature matrix and head-to-head comparisons under matched tasks.

Existing system / literature	Established capability	Potential gap to test, not assume
Nipype (Gorgolewski et al., 2011)	Python interfaces and directed scientific workflows; provenance and diverse executors	Comparative usability for non-programmers, plugin manifest portability, operator-level receipts
Pydra	Task/workflow execution and caching; extensible Python execution	End-user discoverability and interactive QC, not a claim to better underlying scheduling
BIDS / BIDS Apps (Gorgolewski et al., 2016/2017)	Standardized dataset layout and app conventions	Agent discoverability and integrated execution/QC across heterogeneous tools
Boutiques (Glatard et al., 2018)	Machine-readable application descriptions with input constraints and command generation	Scientific metadata, multi-tool provenance, user-facing workflow life cycle
CWL / Nextflow / Snakemake	Workflow definitions, portability, DAG execution and reuse	Neuroimaging-specific on-boarding, validation and clinician-facing results
Neurodesk / Neurodocker	Containerized neuroimaging tools and deployment	Workflow-level orchestration and run-evidence, not image packaging
CBRAIN	Web portal for accessible distributed	Portable local/desktop mode and

	neuroimaging compute	comparison of setup/reproducibility
brainlife.io	Cloud ecosystem for neuroimaging workflows and reproducible data processing	Deployment independence, BIDS-aware plugin/agent contracts
ChRIS	Medical image processing platform and plugin workflow system	Comparative extension model, provenance and deployment ergonomics
Flywheel / QMENTA / LONI Pipeline	Data/workflow management, platform services and/or workflow coordination	Open installable components and validated capability model; avoid invented differentiators
DataLad / ReproNim / BABS	Dataset and analysis versioning, reproducibility and BIDS-App workflows	Run manifests and audit integration; reuse rather than duplicate
MCP and OpenAPI	Agent-facing tools and standardized service interfaces	Machine-readable scientific meaning and execution authorization; addressed in Paper 1

2.1 High-priority literature analysis

Gorgolewski et al. (2011) Nipype: essential baseline for tool interfaces, workflow DAGs and execution. Any architecture diagram should explicitly show what NeuroInsight delegates to mature workflow executors.

Gorgolewski et al. (2016) BIDS: standardized neuroimaging input layout and metadata, underpinning import, validation and derivative handling. BIDS App guidelines should shape pipeline execution conventions.

Gorgolewski et al. (2017) BIDS Apps: provides reproducible tool packaging and execution; the platform should wrap rather than redefine the convention.

Glatard et al. (2018) Boutiques: important machine-readable application-description prior art. Compare schema expressivity, discoverability, command generation and validation side by side.

Neurodesk and container-centric work: important evidence that software availability and isolation have already improved substantially; the contribution must lie elsewhere.

Open-source platforms to investigate analytical flexibility in neuroimaging (Imaging Neuroscience, 2025): direct comparative survey of BIDS Apps, Neurodesk, Boutiques, Nipype, Pydra, CBRAIN, brainlife.io and other systems; use its taxonomy to construct fair comparator selection.

W3C PROV and RO-Crate: use provenance entity/activity/agent relationships and research object packaging instead of proposing an incompatible new provenance vocabulary.

Agentic scientific workflow systems and Paper2Agent (2026): show that LLM-mediated research action is no longer conceptually novel. Paper 5 should present careful implementation and system-level evaluation, not autonomous-science claims.

2.2 Exact gap and originality test

Candidate defensible combination: local-first and HPC-compatible execution + BIDS-native workflow manifests + integrated QC/provenance + human/agent dual interaction + independent deployability. This combination is a hypothesis about practical value, NOT proof of priority or novelty.

Before naming novelty, make a requirements-to-competitor matrix with current tested versions, hands-on benchmark notes, supported features, citations and screenshots. Record negative findings as well as advantages. Consider contributing upstream if no compelling architectural distinction remains.

3. Public tool requirements and architecture

Primary users: graduate students, neuroimaging engineers, computational researchers, research groups without dedicated DevOps, and analysts requiring audit-ready runs. Non-clinical research-only outputs should be labeled clearly.

MVP user journey: load BIDS dataset → validate inputs → choose plugin or workflow → review parameters and permissions → preview resource requirements → execute locally → monitor stages → inspect QC/outputs → export versioned reproducibility bundle.

Architecture suggestion (illustrative, to be verified against current code): React UI; FastAPI orchestration API; PostgreSQL metadata; object/file storage for artifacts; worker queue; Docker/Apptainer execution adapter; Slurm submission adapter; plugin manifests YAML or JSON; BIDS library; provenance recorder; scientific QC engine.

No need to invent orchestration primitives already provided by Pydra, Nextflow, CWL, Nipype or Slurm. Choose a workflow abstraction that minimizes custom executor responsibilities.

Component	First release	Later release
Plugin registry	Versioned manifests, input/output schema, dependency and resource declaration	Federated registry and agent discovery
Workflow builder	Known safe recipes, parameter forms, DAG preview	Agent-assisted workflow proposal with explicit review
BIDS integration	BIDS validator, metadata introspection, derivative organization	Cohort-aware input suggestions
Execution	Local isolated containers; signed/allowlisted images	Approved HPC backend, resumability and advanced scheduling
QC and reporting	Per-stage check status, logs, output inspection, downloadable report	Extended imaging QC modules and reviewer annotation
Reproducibility	Run manifest, hash/version/config/environment and random seeds	RO-Crate export and DataLad integration
Security	RBAC, no shell passthrough, input allowlists, explicit run approval	Fine-grained delegated agent scopes and policy audit

3.1 Example plugin contract (illustrative)

```
name: fastsurfer_anatomical
version: 0.1.0
scientific_purpose: cortical_and_subcortical_segmentation
inputs: {t1w: {format: NIfTI, bids_suffix: T1w}}
outputs: {segmentation: nifti, statistics: tabular, qc: report}
executor: {kind: container, image_digest: REQUIRED}
resources: {cpu: 8, memory_gb: 32, gpu: optional}
checks: [input_schema, stage_completion, output_exists, qc_required]
provenance: [input_hash, software_version, parameters, environment]
```

This is not a real confirmed FastSurfer execution manifest. Exact software options and resource demands must be derived from supported tool versions and validated in integration tests.

3.2 Permissions, privacy and safety

The platform must never automatically transmit restricted MRI or patient data to external model APIs. Local execution and metadata-only operation should be possible. No unrestricted model-written shell commands. Use bounded allowlisted operations, resource quotas, staging workspaces, explicit approvals, data minimization, and provenance audits.

Because biomedical images may contain identifiers and PHI, provide deidentification guidance, restricted storage policies, and a no-PHI public-demo policy. Research-only intended use must be explicit; do not imply clinical diagnostic validity or FDA clearance.

Institutional software, grants, trademarks and employment-related IP must be reviewed before reusing internal NeuroInsight code in external organizational projects.

4. Work packages and release artifacts

WP	Deliverable	Demonstration of completion
WP0	Prior-work matrix and system requirements	Evidence-backed feature comparison; target persona interviews
WP1	Plugin SDK and versioned schema	Five reproducible independently installable plugins; contract tests
WP2	BIDS intake and validation	Public BIDS dataset imported with documented reports
WP3	Local workflow engine	Multi-stage workflow executed and retry/failure tested
WP4	QC/provenance export	Complete run receipt and output comparison script
WP5	Desktop/web interface	New researcher completes a prescribed task
WP6	Approved HPC adapter	Same reference task executed on permitted Slurm environment
WP7	Agent adapter (optional)	Proposed config, human approval, deterministic validation
WP8	Community beta	External installation reports, public issues, tagged release

5. Experimental research plan

Experiments must specify outcomes, datasets, exact software versions and comparator tasks before collecting results. Distinguish software acceptance tests (does it function?) from research evidence (is it better, for whom, under what constraints?).

Primary outcome proposal: successful independently verified workflow completion in a bounded task, measured from starting installation/configuration state through validated output artifacts. Secondary outcomes: setup time, task interventions, error count, reproducibility evidence, total compute and execution cost.

5.1 Datasets and workflow cases

Case	Data	Reference pipeline	Ground truth / QC
A. Anatomical	Small public OpenNeuro BIDS T1w subset	BIDS check → FastSurfer/FreeSurfer → segmentation/statistics	Complete file set, anatomical QC, regional tables
B. Diffusion	Licensed/public DWI BIDS subset with metadata	QSIPrep or appropriate preprocessing → registration → tractography → connectome	Transformation/QC, matrix shape, pipeline provenance
C. Synthetic invalid	Generated BIDS manifests, corrupt/swap fixtures	Schema check → controlled failure path	Expected error, error class, safe stop
D. Reproducibility	Public reference data processed in two supported environments	Same versioned recipe on local and approved HPC	Predefined output equivalence and run receipts

Avoid claiming numerical identity across heterogeneous CPUs/GPUs; define exact byte-level equality only where feasible, and tolerance-based or scientifically equivalent outputs where appropriate. Public dataset licenses, subject counts and modalities must be verified before registration.

5.2 Baselines and fairness

Condition	Comparison purpose	Fairness precautions
Documented CLI + containers	Traditional expert setup	Same tool versions, inputs, machine; detailed script
Nipype or Pydra workflow	Established Python orchestration	Same pipeline stages and processing options
BIDS App run	Standardized application interface	Do not attribute underlying algorithm speed to UI
Existing research portal (CBRAIN/brainlife.io if accessible)	External integrated system	Only compare like-for-like tasks with accessible infrastructure
NeuroInsight manual UI	Platform contribution	No AI confounding
NeuroInsight + agent option	Agent assistive effect	Same executor and validation rules; agent cost tracked

5.3 Core experiments

1. Experiment E1 — Onboarding and use: recruit consenting participants with varying neuroimaging expertise, assign randomized/counterbalanced tasks to NeuroInsight and a comparator. Outcomes: correct completion, setup minutes, interventions, SUS/task-load where suitable. Collect clear consent; request IRB determination when human-subject research is involved.
2. Experiment E2 — Execution reliability: run fixed datasets and pipelines repeatedly under controlled failure injection (missing image, invalid parameter, interrupted worker, insufficient memory, unavailable container). Quantify recovered versus irrecoverable failures, silent failures and debugging time.
3. Experiment E3 — Cross-environment reproducibility: run identical versions and configs locally and on Slurm, then compare input checksums, lineage, output equivalence and run manifests. Record hardware-specific numerical differences.
4. Experiment E4 — Extensibility: independently implement plugins using published SDK, measure developer effort, interface conformity and first-time success. Compare against at least one established plugin description approach (e.g. Boutiques).
5. Experiment E5 — Provenance reconstruction: blinded reviewers attempt to reconstruct runs from provided logs/manifests; measure success, missing details, time to reconstruct and inter-rater agreement.
6. Experiment E6 — Optional agent assistance: compare manual task configuration with agent suggestions using bounded approvals and deterministic checks; measure incorrect submissions, assistance time and human overrides.
7. Experiment E7 — Biological research demonstration: run a focused, pre-specified, appropriately powered imaging study (e.g. regional volume summaries, or a reproducibility analysis in connectomics). Claims must be tied to actual validated results, not a fabricated discovery.

5.4 Experimental controls and a preregistration outline

- Freeze benchmark task definitions and output validity criteria before the final test.
- Keep comparator software versions, algorithm options, input datasets, resources and expertise comparable.
- Separate development data from held-out test subjects and workflow tasks.
- Log number of attempts, agent tokens, API costs, CPU/GPU hours, failures and human corrections.
- Use expert adjudication for ambiguous or clinically relevant QC judgments, with blinding where feasible.

- Use a pilot to estimate variance and task difficulty before sample-size calculation.

6. Statistical analysis and evidence reporting

Binary success: report counts, paired risk differences with confidence intervals, and a suitable paired test or hierarchical logistic regression with participant/task clustering as required. For repeated task attempts, model task and participant effects.

Time/cost: report distributions and confidence intervals; use paired analysis, bootstrapping, or robust mixed models rather than normality assumptions. Prespecify primary outcome and multiple-testing handling.

Reproducibility: report artifact completion, numeric differences or tolerance pass rate for each output type; do not collapse all output differences into one unsupported score.

Adoption: document external installations, validated workflows, support responses and independent use; distinguish telemetry opt-in statistics from inferential experimental samples.

Failures must be described even if unfavorable, including tool incompatibility, version pinning burden, hidden executor errors, compute variability, and UI complexity.

7. Manuscript paths and proposed abstracts

Path A—Software Toolbox: primary claim is that NeuroInsight is a reusable, validated neuroimaging research workbench. This fits Imaging Neuroscience if the software is original, significant, openly available, compared with existing toolboxes and validated with real datasets.

Path B—Computational Biology Software: claim is broad biological utility, exceptional capabilities and demonstration of biological insights. Target PLOS Computational Biology if mature and open with a convincing broad-relevance case.

Path C—Software Engineering Paper: portable deployment, provenance, reliability and extensibility contribute independently of a biological method; SoftwareX, GigaScience or JOSS may fit.

Path D—High-tier Methods: if a new computational method is discovered and validated (rather than just integration), reformulate a separate methodological paper; Nature Methods requires substantial novelty, wide interest, comparative evaluation and important biological application.

7.1 Detailed manuscript outline — Imaging Neuroscience Software Toolbox

8. Title and structured summary: platform problem, design, verification and availability.
9. Introduction: ecosystem and unmet need; explicit comparisons with existing tools.
10. System architecture: plugin interfaces, compute adapters, data flow, QC/provenance, permissions.
11. Materials and methods: dataset selection, software versions, pipeline designs, baseline conditions and evaluation.
12. Results I: independent reproducibility and platform reliability.
13. Results II: usability and extensibility, if supported by controlled tests.
14. Results III: imaging case study and biologically interpretable QC evidence.
15. Discussion: trade-offs, scope limits, dependence on existing packages, security and generalizability.
16. Code/data availability: pinned release, open source license, archived example data and reproducible commands.
17. Supplement: manifests, exact compute profiles, task sheets, additional runs, failures and install tests.

7.2 Manuscript outline — JOSS

18. Summary and statement of need
19. State of the field and build-versus-contribute justification
20. Software design and key architectural tradeoffs
21. Research impact or credible near-term significance
22. AI-use disclosure
23. Repository, archival release and references

7.3 Example abstract scaffold — do not treat placeholders as results

Background: Neuroimaging research workflows rely on heterogeneous tools and compute backends, complicating accessible and reproducible execution. Methods: We developed NeuroInsight, an open modular workbench combining [components]. We compared it with [prespecified approaches] on [datasets/tasks] using [metrics]. Results: [insert measured reproducibility, reliability and user-study estimates with CIs]. Conclusion: [interpret validated findings, limitations and research utility]. The source code and reproduction package are available at [permanent identifiers].

7.4 Proposed figures and tables

- Figure 1: Architecture and trust boundaries (frontend, registry, executor, provenance).
- Figure 2: User workflow and controlled agent/human approval steps.
- Figure 3: Reproducibility protocol across compute environments.
- Figure 4: Performance and failure modes across comparators (real measured data).
- Figure 5: A real-data imaging case study with QC and reproducibility receipts.
- Table 1: Existing platform feature matrix with evidence and versions.
- Table 2: Dataset/workflow inventory and scientific validation criteria.
- Table 3: Experiment designs, sample sizes and statistical endpoints.
- Table 4: Benefits, limitations and failure categories.

7.5 Reviewer objection response plan

Likely criticism	Required answer
This is a GUI for existing tools	Demonstrate tested, significant new integration capabilities, reproducibility or accessibility outcomes.
Why not CBRAIN/brainlife/Nipype?	Fair cited feature matrix, implementation comparison and matched evaluation.
No biological insight	Include meaningful real-data scientific use case and proper interpretation.
Not independently reproducible	Provide frozen example datasets, pinned images, commands and validated outputs.
No external users	Obtain independent installations and documented use or justify credible near-term demand.
Agent design is unsafe	Show permissions, approval, deterministic constraints and safe failure handling.
Clinical claims unwarranted	Maintain RUO scope and explicit non-diagnostic limitations.
Novelty duplicated by papers 1–3	Delineate system/platform contributions and cite shared components.

8. Journal-tier strategy and acceptance gates

Tiers below are qualitative strategic categories, not journal quartiles or acceptance predictions. Always recheck current author instructions and article types immediately before submitting.

Target / tier	Fit	Minimum persuasive evidence
Nature Methods — A+ stretch	Only if a new life-science method beyond platform integration emerges	Major conceptual method, strong comparisons, important biological application, broad adoption potential and deep validation
Nature Computational Science — A+ stretch	Only for a domain-general computational systems contribution	Cross-domain validation, nontrivial computational novelty, strong general baselines
PLOS Computational Biology — A strong	Open-source software with broad computational biology utility	Exceptional capability, compelling biological use, usable source/tests, clear adoption potential
Imaging Neuroscience — A strong, best natural fit	Original neuroimaging Software Toolbox paper	Open and substantial software; real-data results; clear comparator analysis; reproducible demo
GigaScience — strong open-science route	Reusable workflows, open data and computational resource	End-to-end open reproducible infrastructure with substantial scientific utility
SoftwareX — specialist practical	Well-designed software with clear research application	Mature package, documentation, runnable example, demonstrable utility
JOSS — specialist software	Open-source, maintainable, reusable research package	OSI license, tests, open public history >6 months, research impact, AI disclosure

8.1 Journal-specific non-negotiables

Imaging Neuroscience Software Toolbox (official guide): open source and openly available toolbox at submission, sample data, meaningful real-data application, demonstration of effectiveness, discussion of similarities/differences relative to existing software, and validation of novel algorithms. Narrow plugin extensions have lower fit.

PLOS Computational Biology Software (official guide): open-source tool of broad utility that materially helps provide biological insight; broad adoption or credible potential; OSI-compliant anonymous downloadable source; install/build/test instructions; required short software-article structure (<3,500 words excluding supplements under current guide); cover letter justification.

JOSS (official submission/review guidance): sustained public development over more than six months, releases/issues or other activity across time, feature-complete software, good documentation, automated tests, OSI license, impact case and AI usage disclosure.

Nature Methods (official scope): methods should be novel or significantly improve existing approaches and be backed by strong validation, an important biological question and performance comparison. Platform polish alone does not clear the bar.

9. Roadmap, people, resources and risk controls

Phase	Research and engineering objective	Exit gate
0: weeks 1–4	IP/license audit, prior work, product requirements, comparator access	Approved public repo scope and requirements matrix
1: months 2–3	Plugin SDK, BIDS intake, local execution and provenance MVP	Independent public-data reference demo
2: months 4–5	QC, failure recovery, reproducibility packaging, automated tests	Repeated reference runs and fault injection pass

3: months 6–7	Beta, external installer testing, baseline preparation	User task success pilot and usability feedback
4: months 8–10	Main benchmarking and validation, real-data demonstration	Prespecified experiments complete, raw evidence frozen
5: months 11–12+	Independent replication, manuscript, archive and submission	Submission checklist satisfied; journal fit reconfirmed

Team: scientific software lead, neuroimaging methods/QC reviewer, infrastructure/security contributor, statistician/evaluation collaborator, independent beta testers. Clinical validation requires qualified clinical collaborators and appropriate oversight.

Costs: prioritize public datasets and local synthetic fixtures; budget compute for real MRI pipelines separately from agent inference. Establish resource quotas, storage lifecycle, reproducible version pinning and a maintenance owner.

9.1 Risk register

Risk	Mitigation
No demonstrable novelty	Pivot to software/toolbox venue; contribute improvements upstream.
Maintenance burden	Narrow initial plugin surface; versioned compatibility matrix.
Clinical PHI/IP concerns	Use public/synthetic data, institutional approval and clean code ownership.
Workflow engine reinvention	Delegate tasks to existing proven engines.
Agent makes unsupported claims	Require explicit approval; deterministic checks; audit receipts.
Weak user study	Pilot task difficulty; appropriate consent and independent recruitment.
Comparison bias	Pre-register, fix pipelines, publish failures, version competitors.
Data licensing	Validate redistribution and usage terms before bundling.

10. Immediate first public release: tangible minimum

Recommended first integrated public demonstration: download a small public BIDS MRI dataset, validate metadata, run a containerized anatomical processing plugin, inspect input/output/QC, and export a reproducible run package with hashes and software versions. A second demonstration should execute a compact diffusion/connectomics pipeline and clearly identify scientific QC that remains manual.

Developer-facing assets: Python SDK, plugin schema, manifest validator, sample plugin package, API documentation, test fixtures, CI, issues/roadmap, release notes and contribution guide.

User-facing assets: install guide for one supported OS first, datasets/examples, dashboard, task logs, failure explanations, export/download artifacts, privacy statement and RUO disclaimer.

Adoption approach: pilot with two independent groups, publish reproducible example and use-case tutorial, accept external bug reports, tag stable releases and archive DOI. Public telemetry should be opt-in, minimal and never include patient data.

11. Prioritized reading list and traceable references

Links below are intended as a live reading list; bibliographic metadata and versions should be checked and normalized in the final manuscript. Official journal author guides control submission requirements.

[1] Gorgolewski et al. 2011. Nipype: a flexible, lightweight and extensible neuroimaging data processing framework in Python. Front Neuroinform. https://doi.org/10.3389/fninf.2011.00013

[2] Gorgolewski et al. 2016. The Brain Imaging Data Structure. Scientific Data. https://doi.org/10.1038/sdata.2016.44

[3] Gorgolewski et al. 2017. BIDS Apps: Improving ease of use, accessibility and reproducibility. PLOS Comput Biol. https://doi.org/10.1371/journal.pcbi.1005209

[4] Glatard et al. 2018. Boutiques: a flexible framework to integrate command-line applications in computing platforms. GigaScience. https://doi.org/10.1093/gigascience/giy016

[5] Open-source platforms to investigate analytical flexibility in neuroimaging. Imaging Neuroscience, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12330840/

[6] BIDS specifications and BIDS Apps guidelines. https://bids-specification.readthedocs.io/

[7] BIDS validator software. https://github.com/bids-standard/bids-validator

[8] Pydra task/workflow engine. https://pydra.readthedocs.io/

[9] Neurodesk scientific computing environments. https://www.neurodesk.org/

[10] CBRAIN web neuroimaging platform. https://portal.cbrain.mcgill.ca/

[11] brainlife.io open neuroimaging infrastructure. https://brainlife.io/

[12] ChRIS research imaging platform. https://chrisproject.org/

[13] Flywheel platform. https://flywheel.io/

[14] Common Workflow Language user guide. https://www.commonwl.org/user_guide/

[15] Nextflow documentation. https://www.nextflow.io/docs/latest/

[16] Snakemake documentation. https://snakemake.readthedocs.io/

[17] DataLad handbook. https://handbook.datalad.org/

[18] RO-Crate specification. https://www.researchobject.org/ro-crate/specification.html

[19] W3C PROV-O. https://www.w3.org/TR/prov-o/

[20] Model Context Protocol introduction. https://modelcontextprotocol.io/introduction

[21] OpenNeuro dataset portal. https://openneuro.org/

[22] Imaging Neuroscience guide for authors. https://direct.mit.edu/imag/pages/guide_for_authors

[23] PLOS Computational Biology submission guidelines. https://journals.plos.org/ploscompbiol/s/submission-guidelines

[24] JOSS submitting guidance. https://joss.readthedocs.io/en/latest/submitting.html

[25] JOSS review criteria. https://joss.readthedocs.io/en/latest/review_criteria.html

[26] Nature Methods aims and scope. https://www.nature.com/nmeth/submission-guidelines/about/aims

[27] Nature Computational Science journal. https://www.nature.com/natcomputsci/

[28] SoftwareX journal. https://www.sciencedirect.com/journal/softwarex

[29] GigaScience journal. https://academic.oup.com/gigascience

12. Final publication gates

- [ ] IP, licensing, privacy and compute authorization clarified.
- [ ] Feature-comparison matrix grounded in actual released competitor versions.
- [ ] Public versioned repository and reproducible install path.
- [ ] Real example datasets with redistributable terms and expected artifacts.
- [ ] Plugin contract tests, end-to-end tests, error tests and CI.
- [ ] Independent lab replicates at least one end-to-end demonstration.

- [ ] Primary hypotheses, outcome rules and baselines recorded before main experiments.
- [ ] Statistical estimates and failures reported without cherry-picking.
- [ ] Manuscript follows exact target journal category and scope.
- [ ] Repository/archive DOI, data/code availability and AI-use disclosure completed.
- [ ] Results and journal claims reviewed independently before submission.

Bottom line: A mature open NeuroInsight platform can support a strong Software Toolbox or software publication. A Nature Methods-level submission would require discovery and validation of a substantial new scientific method, likely overlapping with but distinguishable from Paper 1 or Paper 2. Scientific and product impact should be measured separately.